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Introduction

Those that own the Kenwood TR751 or TR851 will be aware of the time bomb built into them. Regardless of the way they are treated the pots eventually go open circuit. It's a shame because in every other respect they're a great radio.

It is expected that anyone undertaking this work will have reasonable mechanical skills and a capacity to make simple wiring changes to accommodate the replacement pots.

The original pots are dual ganged pairing VOL with SQL and RIT with RF gain. Due to the difficulty in finding replacements the most logical course of action is to sacrifice the RIT and RF gain functions in preference for squelch and of course volume and use single gang pots for both.

VR4 RF / RIT 10K lin + 10K lin

VR5 VOL / SQL 10K log + 50K lin

The volume control remains where it is and the SQL now takes the place of the RF/RIT pot.

Because of the pot layout some modifications need to be made to the PCBs carrying the pots. This includes cutting tracks, adding jumper wires as well as adding a few SMD resistors to take care of the missing RF gain and RIT pots.

It should be possible to complete the replacement in less than 90 minutes.

Required parts

2 x HK7760 knobs from Jaycar

1 x RP8516 50K B pot from Jaycar

1 x 45037 10K A dual gang switch pot from Rockby Electronics

2 x 4k7 SMD resistors (RIT)

1 x 10K SMD resistor (RF gain)

Note:

The specified pots were easily available in Australia in 2017. As time goes by stocks will certainly be depleted.

Steps

You really should be familiar with the circuit and what you are trying to achieve before you start so spend a little time with the service manual.

Remove top and bottom covers of the radio being careful not to damage the speaker or speaker wires then remove the plastic front panel. The VFO knob is secured by a hex screw under the rubber finger ring.

The metal front panel is mounted on hinges. Remove the top silver Philips screw on each side then just loosen the bottom screw on each side. The panel should slide forward then swing down.

Remove the nut and washer from each pot and slide them back a little until you can remove the molex plug for each board. Then extract the pots with boards. DON'T break the wires.

De-solder the pots from the boards and prepare them for the installation of the new pots. Work carefully as you need those boards for the new pots.

Think about the next steps, run through them in your head and perhaps make notes about how you will tackle this next stage.

Replace VR4 (RF & RIT)

The RF gain pot is replaced by a 10K SMD resistor

The RIT pot is replaced by two 4.7K SMD resistors

The new squelch pot goes in place of the old RF gain pot

You will have to cut some tracks to isolate the pot from J10

The new squelch pot needs to be wired across to J9 on the other board

Replace VR5 (VOL & SQL)

***** Cut 4 to 5 mm off the shaft of the new dual gang pot *****

Cut the pins of the inner gang on the dual gang pot as close to the body as possible

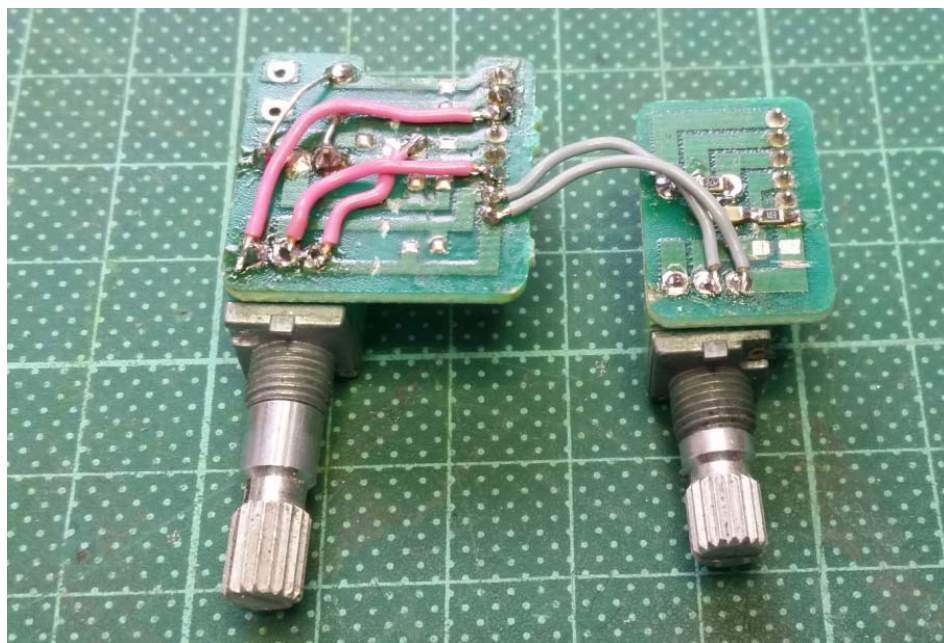
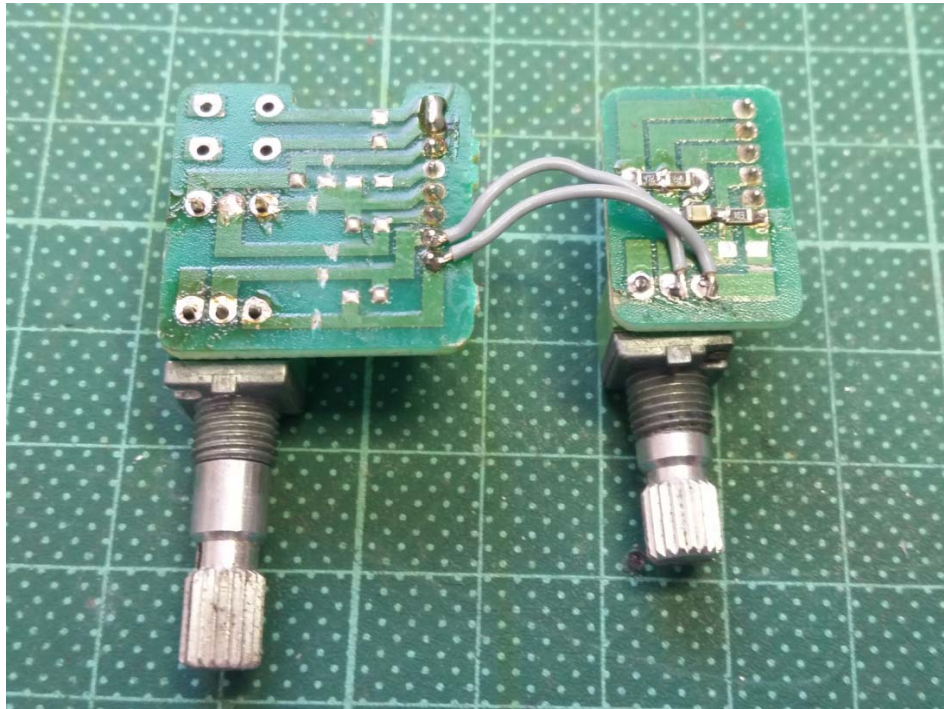
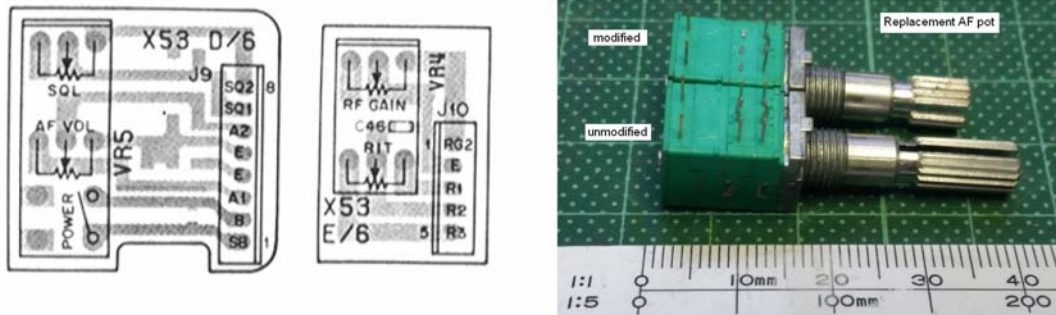
The new VOL/SQL pot will replace the old VOL/SQL pot

The switch contacts will fit in the holes of the old VOL pot

You will have to cut some tracks to isolate the pot from J9

You will have to cut some tracks to isolate the switch contacts from J9

You will have to add a number of jumpers to connect the new VOL pot & switch



Check your work then reassemble the radio but do not install the plastic front panel or covers until you have checked the pots do what they're supposed to do.

If all is well, install the plastic front panel and remaining covers. Take care when re-installing the VFO knob. It is very easy to over tighten the hex screw and split the knob and spares are not available.

Note that the shaft for the replacement SQL pot is rather short. Spread the spline carefully so the knob is a tight fit.

Preparation of the HK7760 knobs

When you look at the recess for the knobs in the plastic front panel you'll see a little lip about 15.5mm in diameter. As luck would have it, exactly the same diameter as the HK7760 knobs. I thought about filing out the recess but instead I put the skirt of each knob on the bench sander, just enough to reduce the diameter so they fits inside the recess. Doing this also means the knobs slide further onto the spline for a better fit.



The modification looks good, certainly not out of place and it's unlikely that a casual observer would even notice the radio is not original.

Disclaimer

This repair is offered as is with no guarantees and with the understanding that it may contain errors.

I assume that most hams will understand the technical details and are capable of building and installing this modification without issue.

Should you break your radio, remember who took the covers off and started this activity. I am not responsible for you breaking your radio. Do not start this repair unless you are sure you are capable of doing the work.